

Three Stable Directions from Quaternionic Minimality: Derivation of $\Sigma_c(n_3) = 3$ from Born–Infeld Fibre Admissibility

O23 — Spectral Admissibility Sub-programme

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Abstract

O21 [28] defined the observable shell n_3 via the threshold condition $\Sigma_c(n_3) = 3$ and used it as structural input for the fibre-level transfer $\delta_{\text{pair}} \rightarrow \beta^*$. O22 established that Born–Infeld projection locking forces saturation to occur on a BFS shell, closing the question “why a shell?” The present paper closes the complementary question left open by O21: why is the relevant saturation threshold three-dimensional, i.e. why $\Sigma_c(n_3) = 3$ rather than 2 or 4? We show that this value is not an arithmetic input but a structural consequence of Born–Infeld fibre admissibility: the value three is not a spectral artefact but the unique dimension of the imaginary subspace of the minimal admissible associative non-commutative algebra. The argument proceeds in three steps. First, an exclusion lemma rules out abelian realizations of the admissible neutral sector and limits the candidate structures to finite subgroups of $\text{SU}(2)$. Second, a quaternionic minimality theorem — anchored in the Hurwitz-to- $\mathfrak{su}(2)$ bridge established in SpectralCapacity [4] — identifies $\mathbb{H}$ as the minimal associative non-commutative structure compatible with Born–Infeld parity, and shows that the admissible neutral directions correspond to the imaginary subspace $\text{Im } \mathbb{H}$, which carries exactly three independent axes. Third, a spectral realization section shows that the ADE binary graphs physically relevant to the programme realize this abstract “3” as exactly three distinct non-trivial spectral levels, confirming the structural derivation at the observable level. The result is conditional on the absence of additional effective symmetries beyond Born–Infeld parity, in accordance with the explicit scope of O18 [25].

Keywords. Cosmochrony; spectral admissibility; quaternionic minimality; Born–Infeld constraint; $\text{SU}(2)$ representations; imaginary quaternions; Weil representation; ADE classification

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1 Position of the Problem after O21–O22

1.1 What O21 established and what it left open

O21 [28] established the shell-level saturation mechanism and derived the fibre-level transfer $\delta_{\text{pair}} \rightarrow \beta^*$. A central ingredient was the threshold condition

$$\Sigma_c(n_3) = 3, \tag{1}$$

which defines the observable shell n_3 used throughout the cascade analysis. O21 used (1) as structural input: the value 3 was interpreted as the number of physically stable directions selected by Π in the Weil realisation of $\text{Heis}_3(\mathbb{Z}/q\mathbb{Z})$, but was not derived from the Born–Infeld structure. O21 §6.1 explicitly identified the derivation of (1) as the open problem of O23.

O22 [29] resolved the complementary question. Theorem 4.8 of O22 shows that projection locking under Born–Infeld spectral admissibility forces saturation to occur on a BFS shell ($n_{\text{sat}} \in \mathbb{N}$), answering “why a shell?” O22 §6.1 confirms that the question “why $\Sigma_c(n_3) = 3$?” thereby becomes well-posed, and assigns it to the present paper. The present result removes the last remaining free structural integer in the spectral admissibility pipeline.

1.2 The three distinct “3”s that must not be conflated

The number 3 appears in three structurally distinct places in the programme, and O23 must keep them separate throughout.

1. $\dim_{\mathbb{R}} \text{Im } \mathbb{H} = 3$: the real dimension of the imaginary quaternions, i.e. the subspace of $\mathbb{H}$ orthogonal to $\mathbb{R} \cdot \mathbf{1}$ and closed under the commutator. This is the algebraic “3” that O23 derives.
2. $\dim_{\mathbb{R}} \mathfrak{su}(2) = 3$: the dimension of the Lie algebra $\mathfrak{su}(2)$ (equivalently, the dimension of $\text{SU}(2)$ as a Lie group). This coincides with (1) because $\mathfrak{su}(2) \cong \text{Im } \mathbb{H}$ as real Lie algebras, but the identification must be derived, not assumed.
3. $\Sigma_c(n_3) = 3$: the observable threshold of O21. This is the spectral “3” that the present paper derives as the signature of (1) at the level of the Gram–Schmidt capacity count.

The causal chain of O23 is: (1) $\Rightarrow$ (2) $\Rightarrow$ (3). Collapsing these three occurrences into a single unexplained “3” would reproduce exactly the gap that O23 is designed to close.

1.3 Scope and what this paper does not claim

O23 derives the value 3 in (1) as a structural consequence of Born–Infeld fibre admissibility. It does *not* determine the numerical value of n_3 as a function of $(c_\chi, \lambda_n, \text{graph geometry})$. That determination requires the extended numerical programme described in O22 §6.2 and remains an open direction. The present result is therefore a structural closure, not a quantitative one.

2 Exclusion Lemma: No Abelian Realisation of the Admissible Neutral Sector

2.1 Setup: neutral generators and the spinorial constraint

We recall the setup of the Admissibility Sub-programme [3, 4]. Let G be a finite group, $S = S^{-1} \subset G$ a symmetric generating set, and $\rho: G \rightarrow \text{SU}(2)$ an irreducible two-dimensional representation. The *neutrality condition* is

$$\chi_\rho(s) := \text{Tr } \rho(s) = 0 \quad \forall s \in S. \tag{2}$$

This condition is the spectral admissibility constraint under bounded relational flux: it selects the generators that remain compatible with the Born–Infeld capacity bound [3, 4]. By the Pauli tracelessness identity, each $\rho(s)$ satisfying (2) can be written

$$\rho(s) = i(\vec{u}_s \cdot \vec{\sigma}), \quad \vec{u}_s \in S^2 \subset \mathbb{R}^3, \quad (3)$$

where $\vec{\sigma} = (\sigma_x, \sigma_y, \sigma_z)$ are the Pauli matrices. Each neutral generator therefore determines a unique axis $\pm \vec{u}_s \in \mathbb{RP}^2$ in the three-sphere [5].

2.2 Exclusion of abelian realizations

Lemma 2.1 (Exclusion of the abelian neutral sector). *No abelian group G admits a symmetric generating set S satisfying the neutrality condition (2) in a faithful irreducible representation of dimension 2.*

Proof. In a one-dimensional representation, $\chi_\rho(s) = \rho(s) \in S^1$, so $|\rho(s)| = 1$ implies $\chi_\rho(s) \neq 0$. Hence faithfulness in dimension 1 is incompatible with (2). In dimension 2, if G is abelian, every irreducible complex representation is one-dimensional by Schur’s lemma over $\mathbb{C}$; a faithful two-dimensional representation of an abelian group decomposes into a direct sum of two one-dimensional characters, and the trace of any element g equals $\chi_1(g) + \chi_2(g)$. Uniform neutrality across a generating set S implies that the trace vanishes on a set generating G . For an abelian group, characters are multiplicative homomorphisms $\chi_i: G \rightarrow \mathbb{C}^\times$; faithfulness of $\rho = \chi_1 \oplus \chi_2$ requires that for every non-identity $g \in G$ there exists i with $\chi_i(g) \neq 1$. The neutrality condition forces $\chi_1(s) + \chi_2(s) = 0$, i.e. $\chi_2(s) = -\chi_1(s)$, for every generator $s \in S$. Since S generates G and characters are multiplicative, this propagates to all of G : $\chi_2 = -\chi_1$ on G . But χ_2 must be a group homomorphism, so $\chi_2(gh) = \chi_2(g)\chi_2(h)$, while $-\chi_1(gh) = -\chi_1(g)\chi_1(h)$. These are equal only if $(-1)^2 = 1$ holds globally, i.e. the sign factor -1 introduced by neutrality must itself be multiplicative. But a global sign $\chi_2 = -\chi_1$ with $|\chi_1(g)| = 1$ for all g is incompatible with multiplicativity unless χ_1 is trivial, contradicting faithfulness. Therefore no abelian group admits a faithful neutral generating set in dimension 2. $\square$

Remark 2.2 (Scope of the exclusion). *Lemma 2.1 excludes all abelian candidates. The remaining candidates are finite non-abelian groups admitting a faithful irreducible two-dimensional representation. By the Klein–Schur classification recalled in [5], these are exactly the finite subgroups of $\mathrm{SU}(2)$: the dicyclic groups Dic_n ($n \geq 2$), the binary tetrahedral group $2T$, the binary octahedral group $2O$, and the binary icosahedral group $2I$. The binary tetrahedral group $2T$ is further structurally excluded by SpectralGramRigidity [5] (its full neutral sector generates only Q_8 , not $2T$). The admissible candidates are therefore Q_8 , Dic_n ($n \geq 3$), $2O$, and $2I$.*

2.3 Exclusion of octonion-type realizations within the Weil framework

Lemma 2.3 (Octonion incompatibility with the Weil realisation). *The octonion algebra $\mathbb{O}$ is incompatible with the class of unitary representations used by the O -series: the Weil representation on $\mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z})$.*

Proof. The Weil representation ρ_c on $\mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ is defined via the cocycle rule

$$\rho_c(g_1 g_2) = \phi(g_1, g_2) \rho_c(g_1) \rho_c(g_2),$$

where ϕ is a $\mathrm{U}(1)$ -valued two-cocycle. Since ρ_c acts on a finite-dimensional complex Hilbert space V via linear operators, its representation matrices are elements of $\mathrm{End}(V)$, which is an associative algebra by definition of operator composition. This construction requires that the algebra of representation matrices be *associative*: the cocycle identity forces $(\rho_c(g_1) \rho_c(g_2)) \rho_c(g_3) = \rho_c(g_1) (\rho_c(g_2) \rho_c(g_3))$. The octonion algebra $\mathbb{O}$ is non-associative: there exist $x, y, z \in \mathbb{O}$ with $(xy)z \neq x(yz)$. A representation taking values in $\mathbb{O}$ cannot satisfy the cocycle identity for all triples (g_1, g_2, g_3) . Hence no Weil-type representation on $\mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ can be realized over $\mathbb{O}$. $\square$

Remark 2.4 (Scope of the octonion exclusion). *Lemma 2.3 does not assert that $\mathbb{O}$ is excluded from all possible physical frameworks. It asserts only that the specific class of Weil representations used throughout the O -series — which requires associativity of the representation algebra via the cocycle identity — is incompatible with an octonionic realization. A broader framework admitting non-associative representations is not excluded in principle; it lies outside the scope of the present programme.*

Corollary 2.5 (Residual classification). *Within the class of unitary Weil representations on $\text{Heis}_3(\mathbb{Z}/q\mathbb{Z})$, the admissible neutral sector is realised by finite subgroups of $\text{SU}(2)$ from the set $\{Q_8, \text{Dic}_n \ (n \geq 3), 2O, 2I\}$. The underlying real associative normed division algebra is necessarily $\mathbb{H}$.*

Proof. By Lemma 2.1 and Remark 2.2, the admissible candidates are finite non-abelian subgroups of $\text{SU}(2)$. By Lemma 2.3, the algebra of representation matrices must be associative. The Hurwitz theorem [30] classifies real associative normed division algebras as $\mathbb{R}$, $\mathbb{C}$, and $\mathbb{H}$. Since $\mathbb{R}$ is one-dimensional (too small for a two-dimensional representation) and $\mathbb{C}$ is commutative (excluded by Lemma 2.1 at the Lie algebra level), the minimal associative non-commutative structure is $\mathbb{H}$. $\square$

3 Quaternionic Minimality and Three Stable Directions

3.1 The Hurwitz-to- $\mathfrak{su}(2)$ bridge (imported result)

The following result is Proposition 7.2 of SpectralCapacity [4], reproduced here for self-containedness.

Proposition 3.1 (Hurwitz-to- $\mathfrak{su}(2)$ bridge, [4]). *Let ρ be an irreducible two-dimensional complex representation of a finite group G , and suppose there exist three linearly independent, mutually non-commuting generators $s_1, s_2, s_3 \in G$ (and their inverses) such that $\chi_\rho(s_i) = 0$ for $i = 1, 2, 3$. Then the matrices $\rho(s_i)$ are traceless elements of $\text{SL}(2, \mathbb{C})$, and the algebra they generate contains a copy of $\mathfrak{su}(2)$ as its real form.*

The proof in [4] proceeds as follows. Each $\rho(s_i) \in \text{SU}(2)$ is traceless, so $\rho(s_i)^2 = -I$. Three mutually non-commuting traceless matrices in $\text{SL}(2, \mathbb{C})$ satisfying the quaternion relations (which follow from their independence and the structure of $\text{SL}(2, \mathbb{C})$) span the Lie algebra $\mathfrak{su}(2) \cong \mathfrak{sp}(1)$ over $\mathbb{R}$. This is the unique associative real structure in real dimension 3, by the Hurwitz classification [30]: among $\mathbb{R}$, $\mathbb{C}$, $\mathbb{H}$, $\mathbb{O}$, the last case is non-associative.

3.2 Neutral directions as axes of $\text{Im } \mathbb{H}$

This subsection mediates the identification between “stable admissible directions” and “generators of $\mathfrak{su}(2)$ ”, a step that must be explicit to avoid conflating algebraic and geometric senses of the word “direction”.

By equation (3), each neutral generator $s \in S$ determines an axis $\pm \vec{u}_s \in \mathbb{RP}^2$. The pairwise character geometry of [5] encodes these axes in the Gram matrix

$$G_{st} = \vec{u}_s \cdot \vec{u}_t = -\frac{1}{2} \chi_\rho(st), \quad (4)$$

which is entirely determined by representation-theoretic data. The rigidity of (4) under admissible deformations means that the relative angles between axes cannot be continuously varied without violating the neutrality condition (2): admissible configurations form discrete families rather than continuous manifolds.

A *stable admissible direction* is therefore an axis $\pm \vec{u}_s$ that belongs to an admissible neutral generating set in the sense of the programme: it is a traceless element of $\text{SU}(2)$ compatible with the Born–Infeld capacity bound. By (3), each such axis corresponds to an element of $\text{Im } \mathbb{H} \cong \mathbb{R}^3$ via $\vec{u}_s \mapsto u_s^x \mathbf{i} + u_s^y \mathbf{j} + u_s^z \mathbf{k}$. The number of independent stable admissible directions is thus the rank of the neutral traceless subspace over $\mathbb{R}$, which equals $\dim_{\mathbb{R}} \text{Im } \mathbb{H} = 3$.

3.3 Main theorem

Theorem 3.2 (Three stable directions from quaternionic minimality). *Let $\Pi: \chi \rightarrow \mathcal{O}$ be a non-injective projection compatible with the Born–Infeld constraint, admitting a Weil realisation on $\text{Heis}_3(\mathbb{Z}/q\mathbb{Z})$. The number of independent stable admissible directions — i.e. the rank of the admissible neutral traceless subspace at the fibre level — is exactly 3. Consequently, $\Sigma_c(n_3) = 3$.*

Proof. By O18 [25], the minimal fibre of Π is the involution $c \leftrightarrow q - c$ (Theorem 3.3 of O18), derived from the evenness of the Born–Infeld action. By Corollary 2.5, the admissible neutral sector is realised within $\mathbb{H}$: the algebra of Weil representation matrices is associative and non-commutative, with $\mathbb{H}$ as the minimal such structure. By Proposition 3.1 (Hurwitz-to- $\mathfrak{su}(2)$ bridge), three mutually non-commuting neutral generators in a two-dimensional irreducible representation span a copy of $\mathfrak{su}(2)$ as their real form. By Section 3.2, the independent stable admissible directions correspond bijectively to independent axes in $\text{Im } \mathbb{H}$. Since $\dim_{\mathbb{R}} \text{Im } \mathbb{H} = 3$ and this subspace is the maximal subspace of $\mathbb{H}$ orthogonal to $\mathbb{R} \cdot \mathbf{1}$ and closed under the commutator, the rank of the admissible neutral traceless subspace is exactly 3. Each independent axis of $\text{Im } \mathbb{H}$ contributes one independent neutral mode to the admissible Gram–Schmidt span; saturation occurs when all three independent axes have been realised by the projection. The threshold condition $\Sigma_c(n_3) = 3$ in O21 [28] records this rank as an observable: the capacity count Σ_c reaches the value 3 precisely at the shell n_3 where all three independent neutral directions have been saturated by the projection. $\square$

Remark 3.3 (Conditionality inherited from O18). *Theorem 3.2 inherits the conditionality of O18 Theorem 2.5: the derivation assumes that Born–Infeld parity is the only global effective symmetry of $S_{\text{BI}}[\chi]$ compatible with BI-admissible observables. If the effective action admits additional symmetries (topological, gauge, or other), the fibre could be larger, and additional neutral directions could be present, modifying the count. A general classification of fibres in enriched frameworks is deferred to a subsequent paper following O21, as noted in O18 §4.3.*

Remark 3.4 (The “3” is not the dimension of $\mathbb{H}$). *The integer 3 derived in Theorem 3.2 is $\dim_{\mathbb{R}} \text{Im } \mathbb{H}$, not $\dim_{\mathbb{R}} \mathbb{H} = 4$. The full quaternion algebra contributes one additional real dimension ($\mathbb{R} \cdot \mathbf{1}$), which is commutative and plays no role in the neutral traceless sector. The Born–Infeld neutrality condition (2) projects out this commutative direction, leaving exactly three independent axes.*

Remark 3.5 (Causal direction: $\text{Im } \mathbb{H}$ precedes $\mathfrak{su}(2)$). *The coincidence $\dim \mathfrak{su}(2) = 3$ is not the origin of the result but a consequence of the quaternionic realisation. The logical order is: Born–Infeld admissibility forces $\mathbb{H}$ -minimality; the imaginary subspace $\text{Im } \mathbb{H} \cong \mathbb{R}^3$ is then the unique neutral traceless subspace; and $\mathfrak{su}(2) \cong \text{Im } \mathbb{H}$ as real Lie algebras follows from the bracket structure of $\text{Im } \mathbb{H}$. Appealing directly to $\dim \mathfrak{su}(2) = 3$ without this chain would be circular: it would assume the $\mathbb{H}$ -structure rather than deriving it.*

4 Spectral Realisation and Consistency with the Programme

4.1 Q_8 as the minimal isotropic prototype

The quaternion group Q_8 is the smallest finite subgroup of $\text{SU}(2)$ admitting a neutral generating set, and SpectralGramRigidity [5] identifies it as the unique admissible group with an orthogonal Gram pattern ($G_{st} = 0$ for all $t \neq s^{\pm 1}$). Its six traceless elements $\{\pm \mathbf{i}, \pm \mathbf{j}, \pm \mathbf{k}\}$ pair into three conjugate pairs, each determining one of the three coordinate axes of $\text{Im } \mathbb{H}$.

Q_8 therefore provides a discrete, exact prototype of the abstract derivation: it realises precisely three independent neutral axes, isotropically distributed, with the second-moment tensor $M \propto I$ [5]. The three neutral axis directions of Q_8 are the finite-group realisation of the three generators of $\mathfrak{su}(2)$, confirming the identification of Section 3.2 in the simplest admissible case.

4.2 Three-level ADE stratigraphic structure

For the ADE binary Cayley graphs physically relevant to the Cosmochrony programme — in particular $2I$ with order-4 and order-5 generators — the Cayley eigenvalue formula gives, for each irreducible representation ρ of G ,

$$\lambda_\rho = |S| - \frac{1}{d_\rho} \sum_{s \in S} \chi_\rho(s).$$

The character sum $\sum_{s \in S} \chi_\rho(s)/d_\rho$ takes exactly three distinct values over the non-trivial irreducible representations of G , corresponding to three distinct eigenvalue classes $\lambda_1 < \lambda_2 < \lambda_3$. This produces exactly three non-zero Cayley eigenvalue classes, and hence three stratigraphic levels [6].

Theorem 3.2 provides the structural explanation for this observation: the three spectral levels are the spectral realisation of the three independent axes of $\text{Im } \mathbb{H}$. The algebraic argument derives the “3” as necessary; the ADE spectral structure shows how this “3” is realised concretely in the relevant substrate graphs. The two arguments are complementary and non-circular: the algebra explains why three is necessary; the ADE confirms how three is achieved.

4.3 Consistency with $\Sigma_c(n_3) = 3$ in O21

O21 [28] defined n_3 as the BFS shell at which Σ_c reaches 3, and O22 showed that projection locking forces saturation onto this shell. Theorem 3.2 now closes the logical chain: the value 3 is the rank of $\text{Im } \mathbb{H}$ as the admissible neutral traceless subspace of the Born–Infeld fibre. The full chain of derivation is:

$$\text{BI parity} \Rightarrow \text{fibre involution } c \leftrightarrow q-c \Rightarrow \text{non-abelian admissible support} \Rightarrow \mathbb{H}\text{-minimality} \Rightarrow \dim \text{Im } \mathbb{H} = 3 \quad (5)$$

5 Discussion

5.1 What O23 derives and what it does not

O23 derives the *value* 3 in the threshold condition $\Sigma_c(n_3) = 3$ as a structural consequence of Born–Infeld fibre admissibility and quaternionic minimality. It does not determine the *location* n_3 as a function of the pipeline parameters (c_χ, λ_n , graph geometry). That question requires the extended numerical campaign described in O22 §6.2 and remains open.

The position of O23 in the O-series is therefore a structural closure of the “why 3?” question, complementary to O22’s structural closure of the “why a shell?” question. Together, O22 and O23 complete the derivation of the observable n_3 at the structural level, leaving the quantitative determination of its numerical value as the remaining open task.

5.2 Conditional status and possible extensions

The main result carries the conditionality of O18 Remark 2.6 (recalled in Remark 3.3 above). If the Born–Infeld effective action admits additional symmetries beyond parity — arising, for instance, from topological constraints identified in the Topological Invariants paper [21] or from gauge structure in an enriched framework — the fibre could be larger and the neutral traceless subspace could have a higher rank. Determining whether such extensions are realised in the Cosmochrony framework, and if so what rank they imply, is deferred to the general fibre classification programme announced in O18 §4.3.

6 Conclusion

We have derived the threshold condition $\Sigma_c(n_3) = 3$ of O21 [28] from Born–Infeld fibre admissibility and quaternionic minimality. The central result is Theorem 3.2: within the class of Weil representations on $\text{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ compatible with Born–Infeld parity, the admissible neutral traceless subspace is isomorphic to $\text{Im } \mathbb{H}$, and its real dimension is exactly 3.

The derivation rests on three pillars: the exclusion of abelian realizations (Lemma 2.1); the incompatibility of octonion-type realizations with the Weil cocycle structure (Lemma 2.3); and the Hurwitz-to- $\mathfrak{su}(2)$ bridge (Proposition 3.1, imported from SpectralCapacity [4]).

The result is conditional on the absence of additional effective symmetries beyond Born–Infeld parity, in accordance with O18 [25]. Subject to this condition, it closes the last structural open problem of O21: the “3” in $\Sigma_c(n_3) = 3$ is not a phenomenological input but *the real dimension of the imaginary quaternions as the minimal admissible non-commutative associative neutral structure forced by Born–Infeld parity*. The emergence of three stable directions is therefore not geometric but algebraic in origin.

Acknowledgements. This paper is part of the Spectral Admissibility Sub-programme of the Cosmochrony framework [1].

A Why Not 2? Why Not 7?

The exclusion argument of Section 2 can be summarised concisely by examining what happens at the other dimensions allowed by the Hurwitz classification.

Why not 2 (complex numbers $\mathbb{C}$)? The algebra $\mathbb{C}$ is a two-dimensional real division algebra, but it is *commutative*. A representation taking values in $\mathbb{C}^\times$ is one-dimensional over $\mathbb{C}$, i.e. abelian, and is excluded by Lemma 2.1. In the language of Section 3.2, the imaginary subspace $\text{Im } \mathbb{C} = i\mathbb{R}$ has real dimension 1: only one neutral axis is available, which is insufficient to carry three independent stable directions.

Why not 7 (imaginary octonions)? The imaginary subspace $\text{Im } \mathbb{O}$ has real dimension 7, which would naively suggest 7 stable directions. However, $\mathbb{O}$ is non-associative, and Lemma 2.3 shows that this is incompatible with the Weil cocycle structure: no representation of $\text{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ can take values in $\mathbb{O}$ while satisfying the cocycle identity. The 7-dimensional imaginary space is therefore inaccessible within the Born–Infeld Weil framework.

Summary. The Hurwitz classification $\{\mathbb{R}, \mathbb{C}, \mathbb{H}, \mathbb{O}\}$ has exactly one entry that is associative and non-commutative: $\mathbb{H}$. Its imaginary subspace has dimension 3. The Born–Infeld neutrality condition selects this unique entry, and the resulting count $\dim_{\mathbb{R}} \text{Im } \mathbb{H} = 3$ is forced.

References

- [1] J. Beau, “Cosmochrony: A Pre-Geometric Relational Framework for Emergent Physics,” white paper, Zenodo (2025). DOI: 10.5281/zenodo.17957509.
- [2] J. Beau, “Emergent Spacetime Geometry from Bounded Relational Relaxation,” companion paper, Zenodo (2026). DOI: 10.5281/zenodo.18407505.
- [3] J. Beau, “Spectral Admissibility under Bounded Flux: Non-Abelian Mode Selection on Cayley Graphs of $SU(2)$ Subgroups,” spectr. adm. subprg. paper, Zenodo (2026). DOI: 10.5281/zenodo.18944985.
- [4] J. Beau, “Spectral Capacity Functional and the Binary-Polyhedral Maximality Conjecture,” spectr. adm. subprg. paper, Zenodo (2026). DOI: 10.5281/zenodo.18945273.
- [5] J. Beau, “Pairwise Character Geometry and the Classification of Neutral Generators in $SU(2)$,” spectr. adm. subprg. paper, Zenodo (2026). DOI: 10.5281/zenodo.18969867.
- [6] J. Beau, “Three Particle Generations and Mass Hierarchy from Spectral Stratigraphy,” spectr. adm. subprg. paper, Zenodo (2026). DOI: 10.5281/zenodo.19040490.
- [7] J. Beau, “Asymptotic Saturation of Projective Resolution: Expander Relaxation Graphs,” spectr. adm. subprg. paper, Zenodo (2026). DOI: 10.5281/zenodo.19056975.
- [8] J. Beau, “Projective Resolution Dynamics on Variable-Valence Relaxation Graphs,” O1 spectr. adm. subprg. paper, Zenodo (2026). DOI: 10.5281/zenodo.19076319.
- [9] J. Beau, “Projective Dynamics and Mass Hierarchy: Hierarchical Amplification via Growing Relational Valence,” O3 spectr. adm. subprg. paper, Zenodo (2026). DOI: 10.5281/zenodo.19076244.
- [10] J. Beau, “Subdiffusive Valence Growth under Bounded Relational Flux: Structural Derivation of the Cascade Exponent β ,” O4 spectr. adm. subprg. paper, Zenodo (2026). DOI: 10.5281/zenodo.19101472.
- [11] J. Beau, “Admissible Frontier Saturation and the Cascade Exponent: A Representation-Theoretic Obstruction and its Matrix-Level Refinement,” O5 spectr. adm. subprg. paper, Zenodo (2026). DOI: 10.5281/zenodo.19120328.
- [12] J. Beau, “Matrix-Level Dynamic Redundancy and the Structural Confinement of the Cascade Exponent β ,” O6 spectr. adm. subprg. paper, Zenodo (2026). DOI: 10.5281/zenodo.19140624.
- [13] J. Beau, “Projective Capacity and the Continuum Limit of Admissible Redundancy,” O7 spectr. adm. subprg. paper, Zenodo (2026). DOI: 10.5281/zenodo.19149773.
- [14] J. Beau, “Three-Step Path Fingerprints on LPS Graphs and the Capacity Growth Exponent,” O8 spectr. adm. subprg. paper, Zenodo (2025). DOI: 10.5281/zenodo.19151769.
- [15] J. Beau, “Projective Capacity Beyond Expanders: Polynomial-Growth Relaxation Graphs and the Capacity Exponent,” O9 spectr. adm. subprg. paper, Zenodo (2026). DOI: 10.5281/zenodo.19154517.
- [16] J. Beau, “Polynomial-Growth Capacity Dynamics at Large q : Volumetric Growth Confirmation, Capacity Decay, and the Weil Decomposition Roadmap,” O10 spectr. adm. subprg. paper, Zenodo (2026). DOI: 10.5281/zenodo.19161074.

- [17] J. Beau, “Weil-Block Projective Capacity on Heisenberg Graphs: Resolving the Representation Obstruction and Extracting the Capacity Exponent,” 011 spectr. adm. subprg. paper, Zenodo (2026). DOI: 10.5281/zenodo.19169987
- [18] J. Beau, “Exact Weil-Block Projective Capacity on Heisenberg Graphs: Resolving the Final Obstruction to δ Extraction,” 012 spectr. adm. subprg. paper, Zenodo (2026). DOI: 10.5281/zenodo.19194798.
- [19] J. Beau, “Asymptotic Stability of Exact Weil-Block Capacity on Heisenberg Graphs: Extended Prime Range, Variance Reduction, and Requalification of the $\delta \rightarrow \beta^*$ Tension,” 013 spectr. adm. subprg. paper, Zenodo (2026). DOI: 10.5281/zenodo.19210831.
- [20] J. Beau, “Observable-Class Mismatch and the Corrected $\delta \mapsto \beta^*$ Relation on Heisenberg Graphs: Block Normalisation, Central-Phase Contribution, and Structural Correspondence,” 014 spectr. adm. subprg. paper, Zenodo (2026). DOI: 10.5281/zenodo.19226272.
- [21] J. Beau, “Topological Invariants of Admissible Configurations: Charge Quantisation, Absence of Magnetic Monopoles, and Projective Chirality in Cosmochrony,” compaignon paper, Zenodo (2026). DOI: 10.5281/zenodo.19282962.
- [22] J. Beau, “Observable-Class Derivation Mismatch and the Exact Growth Equation: From Proxy Capacity to Block-Level Dynamics,” 015 spectr. adm. subprg. paper, Zenodo (2026). DOI: 10.5281/zenodo.19281420.
- [23] J. Beau, “Fibre Admissibility and Resolution of the Capacity Exponent Discrepancy,” 016 spectr. adm. subprg. paper, Zenodo (2026). DOI: 10.5281/zenodo.19298427.
- [24] J. Beau, “Fibre-Level Admissibility from Conjugate Weil Blocks: Structural Derivation of the Pair Observable,” 017 spectr. adm. subprg. paper, Zenodo (2026). DOI: 10.5281/zenodo.19299611
- [25] J. Beau, “Minimal Fibre Structure of the Non-Injective Projection from Born-Infeld Indiscernability: Derivation of the Parity Involution,” 018 spectr. adm. subprg. paper, Zenodo (2026). DOI: 10.5281/zenodo.19321413
- [26] J. Beau, “About Canonical Pair Observables in Weil Blocks: Structure of the Residual Amplitude Factor and Pipeline-Independent Formulation,” 019 spectr. adm. subprg. paper, Zenodo (2026). DOI: 10.5281/zenodo.19323720
- [27] J. Beau, “Projective Persistence and the Physical Sub-Spectrum: A Dynamical Selection Criterion for the Capacity Exponent,” 020 spectr. adm. subprg. paper, Zenodo (2026). DOI: 10.5281/zenodo.19339888
- [28] J. Beau, “Shell-Level Saturation and the Fibre-Level Transfer $\delta_{\text{pair}} \rightarrow \beta^*$,” 021 spectr. adm. subprg. paper, Zenodo (2026). DOI: 10.5281/zenodo.19358961
- [29] J. Beau, “Shell-Alignment from Projection Locking: A Discrete Admissibility Theorem under Born-Infeld Saturation,” 022 spectr. adm. subprg. paper, Zenodo (2026). DOI: 10.5281/zenodo.19367375
- [30] A. Hurwitz, “Über die Composition der quadratischen Formen von beliebig vielen Variablen,” *Nachrichten von der Gesellschaft der Wissenschaften zu Göttingen*, Mathematisch-Physikalische Klasse, pp. 309–316, 1898.